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DRDO & IAF conduct successful flight-tests of RudraM-II Air-to-Surface Missile

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Defence Research & Development Organisation (DRDO) and the Indian Air Force (IAF) have conducted the successful flight-tests of RudraM-II Air-to-Surface Missile from an airborne platform. The tests were conducted under extreme release conditions with critical trajectory establishing the capability of all subsystems.



RUDRAM-II

⇒ chandipur

- RudraM-II is an indigenous air-to-surface missile that was successfully flight-tested by the Defence Research and Development Organisation (DRDO) and the Indian Air Force (IAF). This was tested from Sukhoi-30 Aircraft



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PINAKA LONG RANGE GUIDED ROCKET (LRGR)

- Defence Research & Development Organisation (DRDO) conducted a successful flight-test of Pinaka Long Range Guided Rocket (LRGR) at the Integrated Test Range (ITR), Chandipur on July 08, 2026.
- The rocket was tested for a user defined minimum range of 60 kms.



HQ - China

Israel

David Slings
Arrow

Barak

Iron Dome

Python

Spike

Spyder

LoRa

Harop

USA

THAAD
SHORAD

PATRIOT

HARPoon

Hwasong
+ N.K.

Russia

① Sarmat

② Yars

③ Bulava

④ Iskander

⑤ Zircon

⑥ Oniks

Iran

Bavar
Fattah

Pak
↓

Ghawii

France

Scalp



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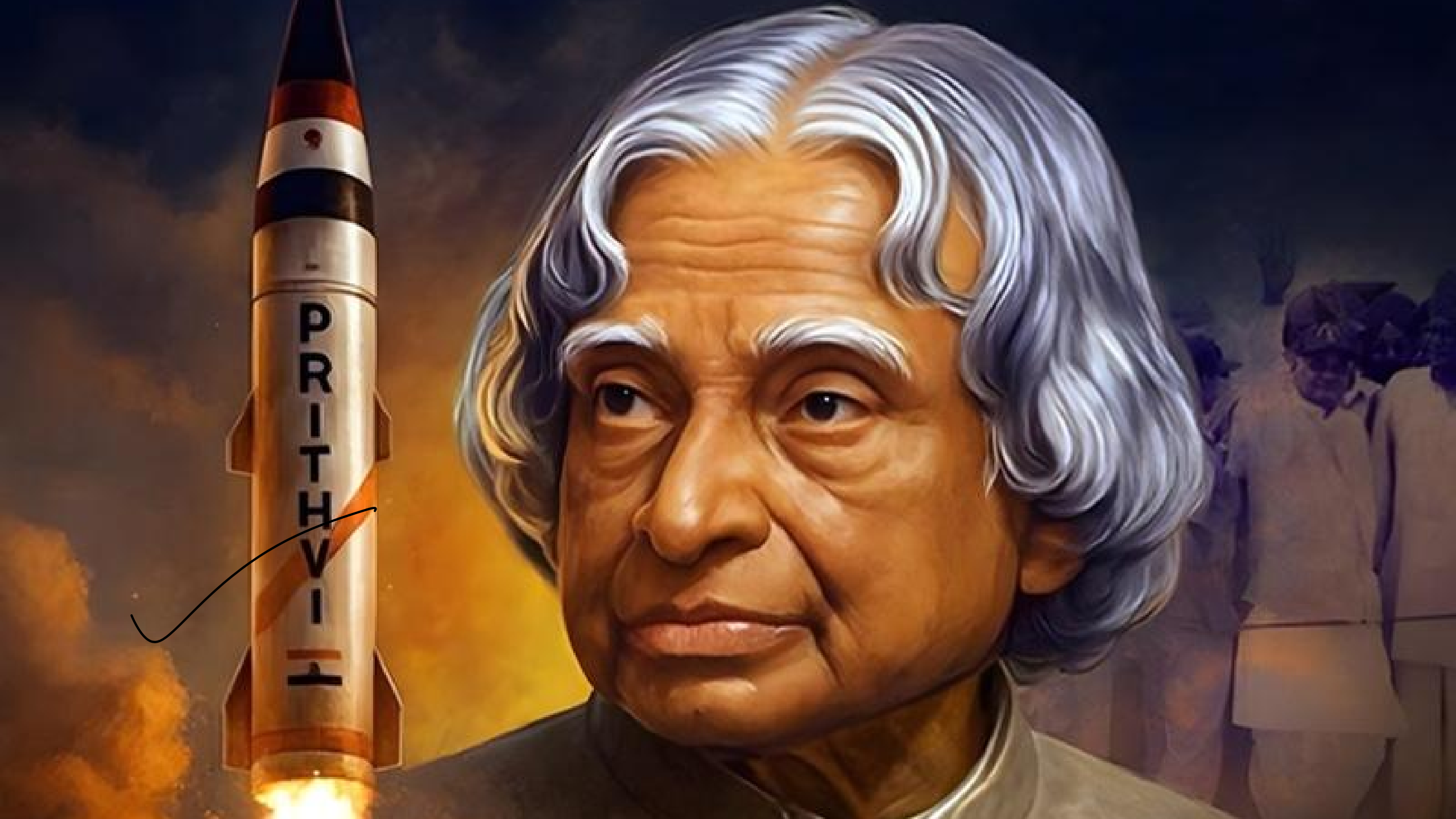


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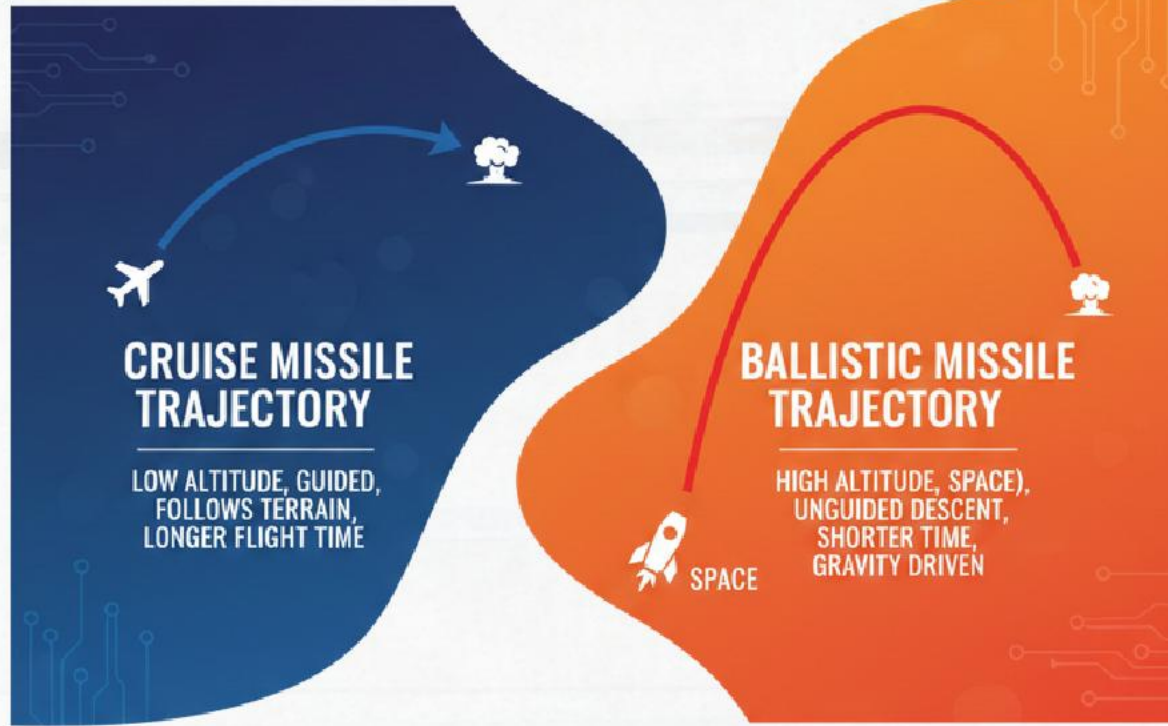
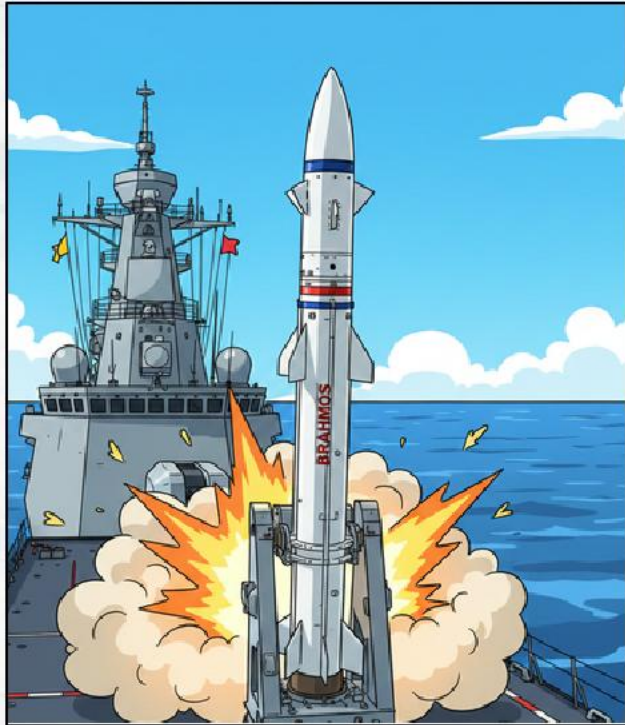


INTEGRATED GUIDED MISSILE DEVELOPMENT PROGRAMME

- **IGMDP (5 missiles, 1983, led by Dr. Kalam, completed 2008).**
- **Missiles Developed:**
- **Prithvi – Short-range (150–600 km) surface-to-surface ballistic missile.**
- **Agni – Initially tech demonstrator → now family of ballistic missiles (Agni-I to Agni-V).**
- **Trishul – Short-range surface-to-air missile.**
- **Nag – 3rd generation anti-tank guided missile.**
- **Akash – Medium-range surface-to-air missile.**



CRUISE MISSILES VS BALLISTIC MISSILES





CRUISE MISSILES

- Unmanned, guided, jet-engine powered; fly within Earth's atmosphere using aerodynamic lift (like airplanes).
- Short-range (300–400 km), high precision.
- Types (by speed):
 1. Subsonic ($< \text{Mach } 1$, $\sim 0.8 \text{ Mach}$)
 2. Supersonic ($2\text{--}3 \text{ Mach}$)
 3. Hypersonic ($> \text{Mach } 5$)
- Example: BrahMos.



Agni
Prithvi } Ballistic

CRUISE MISSILE



BRAMHOS II HYPERSONIC



NIRBHAY-SUBSONIC LAND ATTACK

BALLISTIC MISSILES

- Follow a ballistic trajectory: exit Earth's atmosphere, re-enter to hit target.
- Long-range capability; carry heavy payloads (e.g., nuclear warheads).
- Classified by range (SRBM, MRBM, IRBM, ICBM).
- India: Agni series





SUBSONIC MISSILES



- These are missiles that travel at a speed slower than the speed of sound (Mach 1).
- They are designed for short- to medium-range operations with high precision.



SUPERSONIC MISSILES

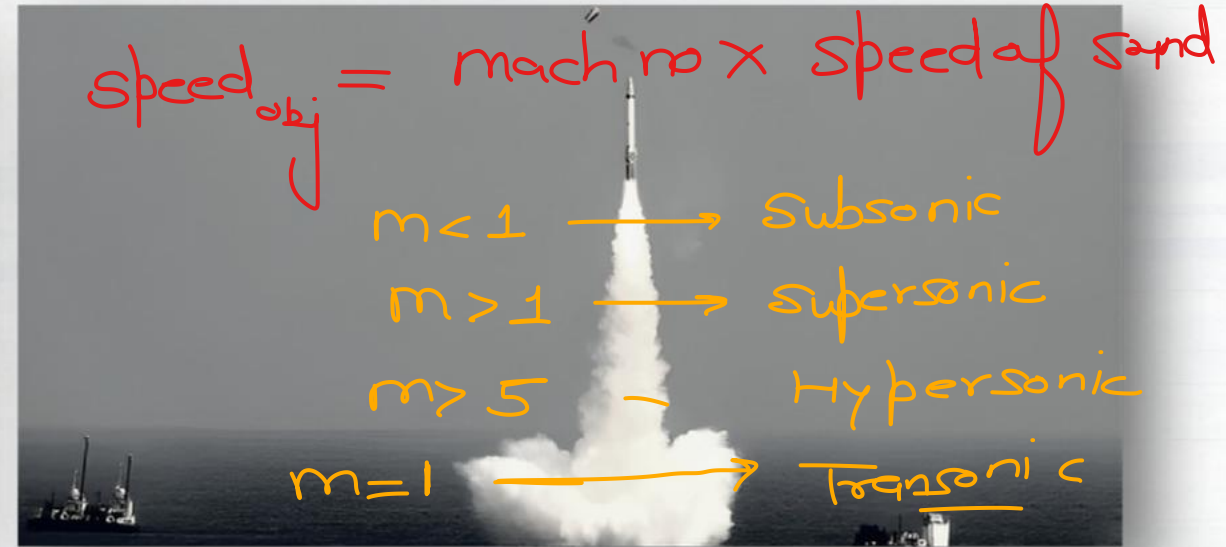


- Missiles that travel faster than the speed of sound (above Mach 1) but below Mach 5 fall into this category.
- They strike targets with greater speed and reduced reaction time. Ex: Bramhos



$$\Rightarrow \text{mach no} = \frac{\text{speed of obj}}{\text{Sound}}$$

HYPERSONIC MISSILES



- The term hypersonic refers to missiles capable of travelling at speeds exceeding Mach 5, i.e., more than five times the speed of sound.
- Extremely fast, making them difficult to intercept. Example: Shaurya



SURFACE TO AIR MISSILES

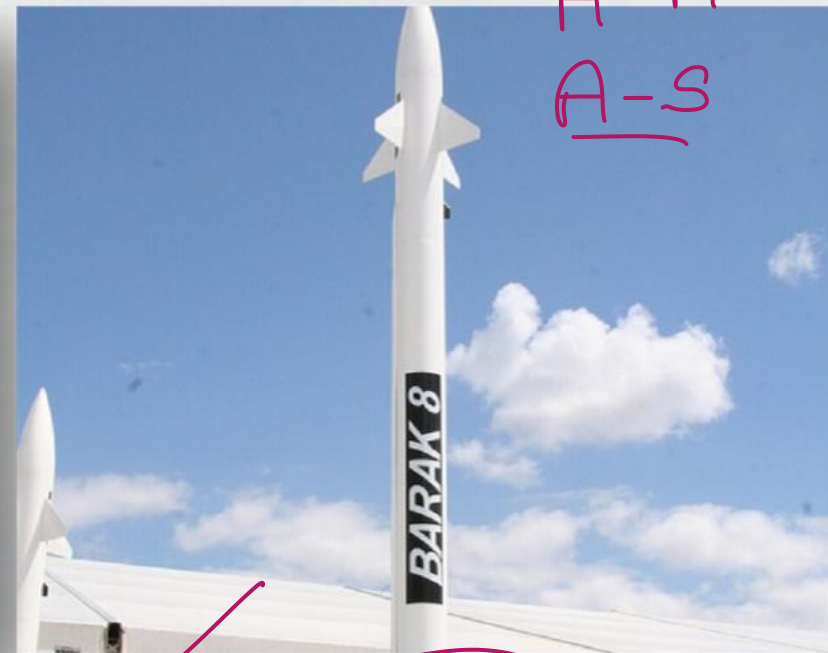
S-A
A-A
A-S



TRISHUL



AKASH



BARAK



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Surf- **SURFACE TO AIR MISSILES**



AGNI



PRITHVI



PRAHAAR



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AIR TO SURFACE MISSILES



RUDRAM



RAMPAGE

Israel



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AIR TO AIR MISSILES



ASTRA



PYTHON



MICA



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Harpoon Anti ship.
missile

Spike
Israel

ANTI TANK GUIDED MISSILE



AMOGHA

Melina



NAG



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ANTI TANK GUIDED MISSILE



HELINA



SAMHO



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AFCAT PYQ BASED

Consider the following sentences

1. The Aeronautical Development Establishment (ADE), located in Bengaluru, is the Nodal Laboratory for the LRLACM project.
2. The missile is an indigenously developed system where all sub-systems have been created by various DRDO laboratories and Indian Industry partners.
3. The flight-test was conducted to verify that all test objectives were met, which was confirmed by data from tracking instruments at the Integrated Test Range (ITR), Andhra Pradesh.

Which of the following is/are correct?

1. 1 and 3
2. 1 and 2
3. Only 1
4. Only 3



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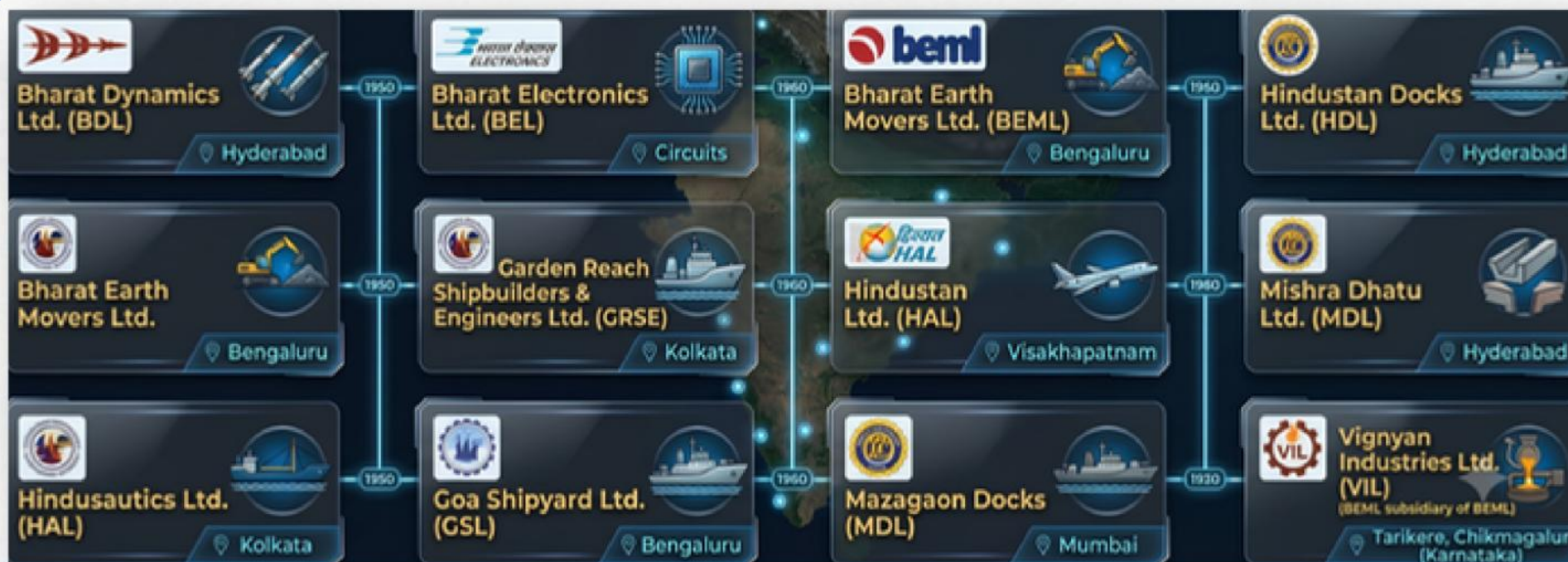
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Disinvestment

MAJOR DEFENCE PUBLIC SECTOR UNDERTAKINGS (PSUS) OF INDIA



THE DEFENCE MINISTRY 5 DEPARTMENTS



DEPARTMENT OF DEFENCE



DEPARTMENT OF DEFENCE
PRODUCTION



DEPARTMENT OF EX-
SERVICEMEN WELFARE (DESW)



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THE DEFENCE MINISTRY 5 DEPARTMENTS



DEPARTMENT OF MILITARY AFFAIRS (DMA)



DEFENCE RESEARCH AND DEVELOPMENT ORGANISATION

Def. maj.

- ① Army
- ② Navy
- ③ A.F.
- ④ ICG
- ⑤ Dircc. of Def. Estate
- ⑥ BRO
- ⑦ DG NCC
- ⑧ CSD
- ⑨ milit. Eng. Ser.

- ⑩ Hydrographic Survey
- ⑪ Off. of Joint Sec

- ⑫ Control. Genl. Def. Acc
- ⑬ DG AFMC
- ⑭ Territorial Army etc.

Assam Rifles

Admis. by MHA

oper. control

↓
Army

70-80%.

friends of Hill people

MHA

Trick - HIS JOB

Dpt of Home
" " Inten. security
" " state
" " J&K
" " Border man
" " Official

INTELLIGENCE AGENCIES

RAW



ISI



MIB



CIA



KGB



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COMMANDS OF INDIAN ARMY



IMPORTANT INDIAN ARMY CHIEFS



SAM MANEKSHAW

- Chief of Army Staff from 1969 to 1973 during tensions with Pakistan and internal conflicts.
- Formulated military strategy for the 1971 Indo-Pakistani War, resulting in the surrender of 93,000 Pakistani troops and the formation of Bangladesh.



1971 - war
1st 2 Shimla agree.



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K. M. CARIAPPA

- First Indian Commander-in-Chief of the Indian Army (1949), taking over from British General Sir Francis Butcher.
- Played a key role in the Indo-Pakistani War of 1947–48 (Kashmir conflict).
- Known for introducing the principle of “Indianisation” of the Army (greater role for Indian officers).



K. M. CARIAPPA

- Indian Army Day is celebrated on 15 January every year because on this day in 1949, General K. M. Cariappa took over as the first Indian Commander-in-Chief of the Indian Army, replacing General Sir Francis Butcher, the last British Commander-in-Chief



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✓ AFCAT PYQ

With reference to Flying Officer Nirmal Jit Singh Sekhon and the Param Vir Chakra, consider the following statements.

1. He was the first individual from the Indian Air Force to be awarded India's highest gallantry award, the Param Vir Chakra.
2. The Param Vir Chakra is octagonal in shape, made of silver, and bears a single Vajra with the Ashoka Chakra embossed at the centre.

Which of the statements given above is/are correct?

1. 2 only
2. Neither 1 nor 2
3. Both 1 and 2
- ✓ 4. 1 only



AFCAT PYQ BASED

5 star rank.

With reference to Field Marshal Sam Manekshaw, consider the following statements.

- 1. He was the first Indian Army officer to be promoted to the rank of Field Marshal.**
- 2. He served as the Chief of Army Staff from 1969 to 1973 during a period of tensions with Pakistan and internal conflicts.**

Which of the statements given above is/are correct?

- 1. 1 only**
- 2. 2 only**
- 3. Both 1 and 2**
- 4. Neither 1 nor 2**



AFCAT PYQ BASED

With reference to the transfer of command of the Indian Army in 1949, consider the following statements.

- 1. General K. M. Cariappa took over as the first Indian Commander-in-Chief of the Indian Army from General Sir Francis Butcher.**
- 2. General Sir Francis Butcher was the last British Commander-in-Chief of the Indian Army.**
- 3. Indian Army Day is celebrated every year on 15 January.**

Which of the statements given above is/are correct?

- 1. 1 only**
- 2. 1 and 3 only**
- 3. 3 only**
- 4. 1, 2 and 3**



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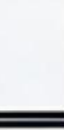
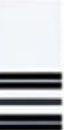


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RANK STRUCTURE - INDIAN ARMED FORCES

COMMISSIONED RANKS									
Indian Army									
									
LIEUTENANT	CAPTAIN	MAJOR	LIEUTENANT COLONEL	COLONEL	BRIGADIER	MAJOR GENERAL	LIEUTENANT GENERAL	GENERAL	FIELD MARSHAL
Indian Air Force									
									
FLYING OFFICER	FLIGHT LIEUTENANT	SQUADRON LEADER	WING COMMANDER	GROUP CAPTAIN	AIR COMMODORE	AIR VICE MARSHAL	AIR MARSHAL	AIR CHIEF MARSHAL	MARSHAL OF THE AIR FORCE
Indian Navy									
									
SUB LIEUTENANT	LIEUTENANT	LIEUTENANT COMMANDER	COMMANDER	CAPTAIN	COMMODORE	REAR ADMIRAL	VICE ADMIRAL	ADMIRAL	ADMIRAL OF THE FLEET

Equiv.



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EQUIVALENT RANKS OF JCO'S

		
NAIB SUBEDAR	SUBEDAR	SUBEDAR MAJOR
		
JUNIOR WARRANT OFFICER	WARRANT OFFICER	MASTER WARRANT OFFICER
		
CHIEF PETTY OFFICER	MCPO SECOND CLASS	MCPO FIRST CLASS



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EQUIVALENT RANKS OF NCO'S

ARMY

NO INSIGNIA

SEPOY



Lance Naik



NAIK



HAVILDAR

AIRFORCE



AIRCRAFTSMAN



LEADING AIRCRAFTSMAN



CORPORAL



SERGEANT

NAVY

NO INSIGNIA

SEAMAN 2ND CLASS

NO INSIGNIA

SEAMAN 1ST CLASS



LEADING RATE



PETTY OFFICER



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v. imp.

clg of military engin. → Pune

MILITARY TRAINING INSTITUTIONS

National Def
clg → ~~Delhi~~ ~~clg~~



INFANTRY SCHOOL (MHOW AND BELGAUM)

High Altitude
warfare schl → Gulmarg



SCHOOL OF ARTILLERY
DEOLALI, MAHARASHTRA



MILITARY TRAINING INSTITUTIONS



COLLEGE OF MILITARY ENGINEERING
PUNE, MAHARASHTRA



MILITARY COLLEGE OF ELECTRONICS AND
MECHANICAL ENGINEERING, SECUNDERABAD



MILITARY TRAINING INSTITUTIONS



NATIONAL DEFENCE COLLEGE
DELHI



DEFENCE SERVICES STAFF COLLEGE
WELLINGTON, TAMIL NADU



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MILITARY TRAINING INSTITUTIONS



**MILITARY COLLEGE OF
TELECOMMUNICATION ENGINEERING, MHOW**



**INSTITUTE FOR DEFENCE STUDIES &
ANALYSIS, NEW DELHI**